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Last time we developed this relationship between the relative error in a computed solution $\bar{x}$ and its relative residual.

$$\frac{\|\bar{\underline{x}} - \underline{x}^*\|}{\|\underline{x}^*\|} \leq \text{cond}(A) \frac{\|\underline{b} - A\bar{\underline{x}}\|}{\|\underline{b}\|}$$

where $\text{cond}(A) = \|A^{-1}\| \, \|A\|$

is the condition number of A - a measure of the sensitivity of the solution to small inaccuracies

Here we'll use induced matrix norms

$$\|A\| = \max_{\|\underline{x}\|=1} \|A\underline{x}\|$$

and we considered a problem with

$$A = \begin{bmatrix} 0.780 & 0.563 \\ 0.913 & 0.659 \end{bmatrix}$$

... norms continued.

$$\|A\|_1 = \max_{\|\underline{x}\|_1=1} \|A\underline{x}\|_1$$

you can show $= \max_{1\le j\le n} \left(\sum_{i=1}^{n} |a_{ij}| \right)$

= max absolute column sum

$$\|A\|_2 = \max_{\|\underline{x}\|_2=1} \|A\underline{x}\|_2$$

you can show $= \sqrt{\text{largest evalue of } A^TA}$

$$\|A\|_\infty = \max_{\|\underline{x}\|_\infty=1} \|A\underline{x}\|_\infty$$

you can show $= \max_{1\le i\le n} \left(\sum_{j=1}^{n} |a_{ij}| \right)$

= max absolute row sum

$$A = \begin{bmatrix} 0.780 & 0.563 \\ 0.913 & -0.659 \end{bmatrix}$$

Also the Frobenius norm

$$\|A\|_F = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}^2}$$

$$\|A\|_1 = \max\left\{ |0.780| + |0.913|,\ |0.563| + |-0.659| \right\} = 1.693$$

$$\|A\|_\infty = \max\left\{ |0.780| + |0.563|,\ |0.913| + |-0.659| \right\} = 1.572$$

$$\|A\|_2 = 1.2159$$

$$\|A\|_F \doteq 1.481$$

back to our example.

$$A = \begin{bmatrix} .780 & .563 \\ .913 & .659 \end{bmatrix} \qquad \|A\|_\infty = 1.572$$

$$A^{-1} = \frac{1}{(0.780 \times 0.659 - 0.913 \times 0.563)} \begin{bmatrix} 0.659 & -0.563 \\ -0.913 & 0.780 \end{bmatrix}$$

det(A) $\rightarrow$

$$= 1 \times 10^{-6} \qquad \|A^{-1}\|_\infty = 1.693 \times 10^{6}$$

$$\therefore \ \text{cond}_\infty(A) = \|A\|_\infty \|A^{-1}\|_\infty$$
$$= (1.572)(1.693 \times 10^6)$$
$$\doteq 2.66 \times 10^6$$

Solving <u>this</u> $A\underline{x} = \underline{b}$ problem with this A is said to be ill-conditioned because we know

relative error in computed solution $\leq (2.66 \times 10^6)$ (relative residual in computed solution)

could be as large as

this being $\neq 0$ gets magnified.

How small can cond(A) be in general?

$$\text{cond}(A) = \|A\| \|A^{-1}\|$$
$$\geq \|A A^{-1}\|$$
$$= \|I\|$$

$$= \max_{\|\underline{x}\|=1} \|I\underline{x}\|$$
$$= \max_{\|\underline{x}\|=1} \|\underline{x}\|$$
$$= 1$$

$\therefore$ cond(A) ≥ 1.

✓ • can solve $A\underline{x} = \underline{b}$

✓ • understand error bound.

Are we done? Not quite

Consider solving the problem:

$$A \rightarrow \begin{pmatrix} 1.0\times10^{-4} & 1 \\ 1 & 1 \end{pmatrix} \overset{\underline{x}}{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}} = \overset{\underline{b}}{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}$$

With <u>exact</u> LU factorization

$$\begin{pmatrix} 1 & 0 \\ 1\times10^{4} & 1 \end{pmatrix}\begin{pmatrix} 1\times10^{-4} & 1 \\ 0 & -9{,}999 \end{pmatrix}$$

Forward solve $\quad L\underline{y} = \underline{b} \quad \begin{pmatrix} 1 & 0 \\ 1\times10^{4} & 1 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$y_1 = 1$

$y_2 = -9998$

Back solve $\quad U\underline{x} = \underline{y} \quad \begin{pmatrix} 1\times10^{-4} & 1 \\ 0 & -9{,}999 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -9998 \end{pmatrix}$

get $\quad \underline{x} = \begin{pmatrix} 10{,}000/9{,}999 \\ 9{,}998/9{,}999 \end{pmatrix}$

To explore the effect of finite precision on calculation, let's repeat the solving using 3 decimal digits with proper rounding.

get $\quad \bar{L} = \begin{pmatrix} 1 & 0 \\ 1.00\times10^{4} & 1 \end{pmatrix} \quad \bar{U} = \begin{pmatrix} 1.00\times10^{-4} & 1 \\ 0 & 1.00\times10^{4} \end{pmatrix}$

$\boxed{\text{exact}}$

$= L$

1 rounding error

Forward subst. $\begin{pmatrix} 1 & 0 \\ 1\times10^{4} & 1 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1.00 \\ fl\left(\frac{2 - 1\times10^{4}}{1}\right) = -1.00\times10^{4} \end{pmatrix}$$

Back subst $\begin{pmatrix} 1.00\times10^{-4} & 1.00 \\ 0 & -1.00\times10^{4} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1.00 \\ -1.00\times10^{4} \end{pmatrix}$

gives $\underline{x} = \begin{pmatrix} 0.00 \\ 1.00 \end{pmatrix}$ ← inaccurate! ← accurate

We got a solution component with a large relative error! Last time, the issue was with ill-cond^{n}

Is A ill-conditioned? $\|A\|_\infty = 2$

$$A^{-1} = \frac{1}{\det(A)}\begin{pmatrix} 1 & -1 \\ -1 & 1\times10^{-4} \end{pmatrix}$$

$1\times10^{-4}-1$ → so $\|A^{-1}\|_\infty = \left|\frac{1}{1\times10^{-4}-1}\right|\cdot 2 \doteq 2$

so $\text{cond}_\infty(A) \doteq 4$.

The effect of a nonzero relative residual will be magnified by a factor of ≈ 4. This is a well-conditioned problem!

The large error is not due to ill-conditioning.

Let's go back and look at the calculations again:

$$\begin{pmatrix} 1 & 0 \\ 1\times10^4 & 1 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \rightarrow \begin{array}{l} y_1 = 1.00 \\ y_2 = -1.00\times10^4 \end{array}$$

contains rounding error of size 2.

$$\begin{pmatrix} 1\times10^{-4} & 1 \\ 0 & -1.00\times10^4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1.00 \\ -1.00\times10^4 \end{pmatrix}$$

$$x_2 = 1.00 \qquad \text{true } x_1 = \frac{9998}{9999}$$

this should read : (1.00 - 1.00 x 1.00) / 1.00 x 10^(-4) r.e. of $1/9999$.

$$x_1 = \left(1.00 - 1.00\times1.00\right) / 1.00\times10^{-4}$$

this cancellation makes this error significant in the result!

We can trace this significant rounding error back to the elimination in the $\underline{b}$ vector where we had

$$2 - 1*1\times10^4$$

this value could have been quite different and not changed the result due to precision limit.

we essentially lose the $\underline{b}$ info.

We lost information because the multiplier was so large (because the value on the diagonal was quite small)